LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Most-Expected Questions: Physics (Ch 9–12)

Chapter / Topic: Light · Human Eye · Electricity · Magnetic Effects of Electric Current

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. These are the question types that come up again and again in board papers. Answer them in your notebook, then check with the answer key.
2. Numericals: given → formula → substitution (with signs) → answer with unit.

1-MARK QUESTIONS

Q1. What is the SI unit of the power of a lens?

Q2. What is the focal length of a plane mirror?

Q3. Which spherical mirror always forms a virtual, erect and diminished image?

Q4. What is the far point of a normal human eye?

Q5. Which phenomenon makes the clear sky look blue?

Q6. Write the SI unit of resistivity.

Q7. Which has more resistance: a 100 W or a 60 W bulb (both rated 220 V)?

Q8. What is the colour of the insulation on the earth wire?

2-MARK QUESTIONS

Q9. The magnification produced by a plane mirror is +1. What does this mean?

Q10. Find the focal length of a lens of power –2.0 D. What type of lens is it?

Q11. You want an erect image of an object using a concave mirror of focal length 15 cm. Where should the object be? Will the image be larger or smaller?

Q12. Why are copper and aluminium wires usually used for transmitting electricity?

Q13. How does the resistance of a wire change when (a) its length is doubled and (b) its area of cross-section is doubled?

Q14. Why does a ray of light bend when it passes from air into glass?

Q15. Why is the tungsten filament of a bulb able to glow without melting? Why are bulbs filled with nitrogen or argon?

Q16. Why is the magnetic field inside a long solenoid said to be uniform?

3-MARK QUESTIONS

Q17. An object 4 cm high is placed 25 cm in front of a concave mirror of focal length 15 cm. Find the position, size and nature of the image.

Q18. An object is placed 12 cm in front of a convex mirror of radius of curvature 30 cm. Find the position and nature of the image.

Q19. A 6 Ω and a 3 Ω resistor are joined in parallel, and this combination is connected in series with a 2 Ω and a 3 Ω resistor across a 12 V battery. Find the total resistance and the current.

Q20. An electric motor takes 5 A from a 220 V line. Find its power and the energy it uses in 2 hours.

Q21. The hot plate of an electric oven on a 220 V line has two coils A and B of 24 Ω each, which can be used separately, in series or in parallel. Find the current in each case.

Q22. A copper wire 1 km long has a cross-sectional area of 0.5 mm^2. Find its resistance. (ρ of copper = 1.62×10^{-8} Ω m)

5-MARK QUESTIONS

Q23. (a) How would you connect three resistors of 6 Ω each to get a total resistance of (i) 9 Ω and (ii) 4 Ω? [3]
(b) Several 10 W bulbs are designed for a 220 V line. How many can be connected in parallel if the maximum allowed current is 5 A? [2]

Q24. (a) Draw ray diagrams for a convex lens with the object (i) at $2F_1$ and (ii) between F_1 and O. State the nature and position of the image in each case. [3]
(b) Why does a convex lens act as a magnifying glass? [2]

Q25. (a) What is myopia? Explain its causes and correction. [3]
(b) Why does the Sun appear reddish at sunrise and sunset? [2]

~ Every numerical: units first, signs second, arithmetic last! ~